

Consumer Attention Mapping System

1. Title: Consumer Attention Mapping System

2. Objective

Build an AI-powered Consumer Attention Mapping System that uses cameras, computer vision, behavioral analytics, and AI models to understand how shoppers interact with retail shelves, products, promotional displays, and store layouts.

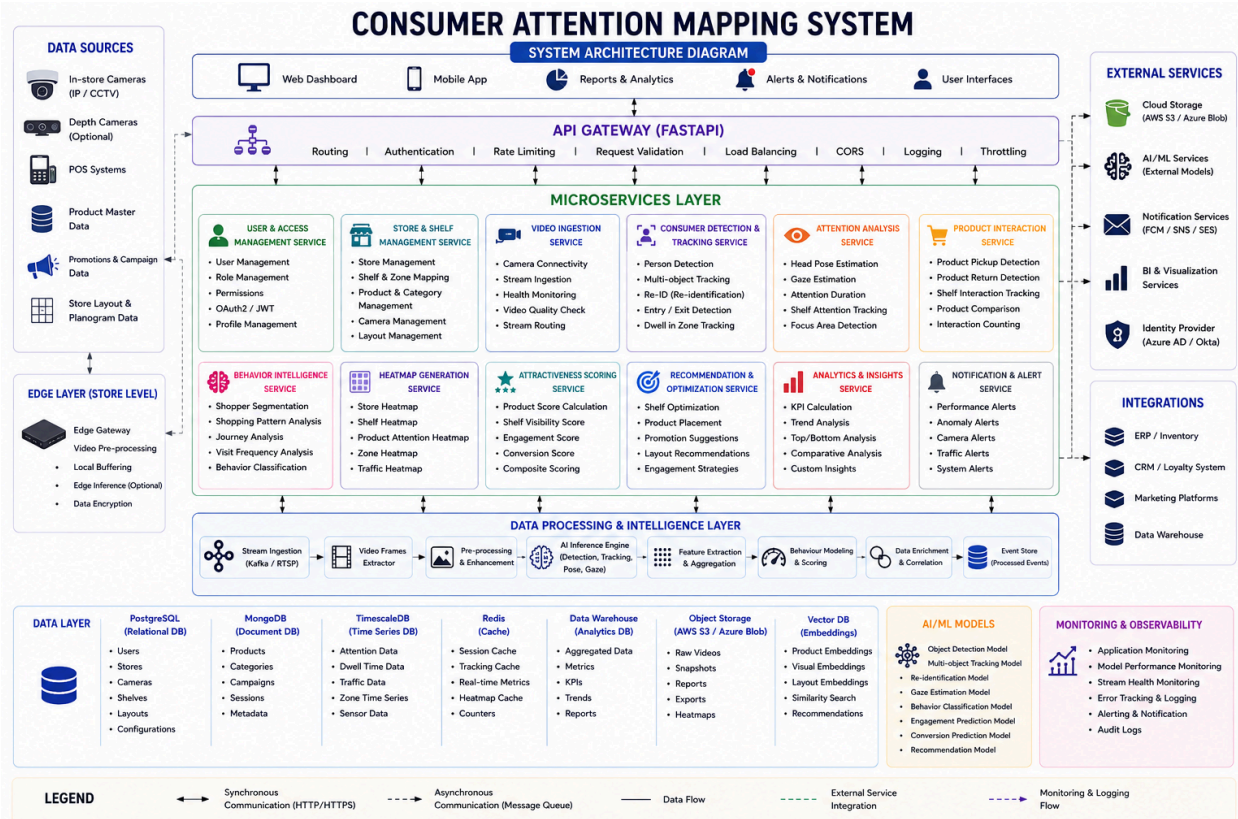
The platform analyzes customer attention patterns, gaze direction, dwell time, product engagement, movement paths, and purchase interactions to help retailers optimize shelf placement, improve product visibility, increase conversion rates, and enhance in-store customer experiences.

The solution is designed for retail stores, supermarkets, shopping malls, FMCG companies, consumer brands, retail analytics teams, and marketing organizations.

Outcomes

- Developed and deployed an AI-powered retail attention intelligence platform.
 - Implemented secure authentication and role-based access control.
 - Built shopper tracking and attention analysis workflows.
 - Developed product interaction and shelf engagement analytics.
 - Implemented consumer behavior modeling and attention heatmaps.
 - Built conversion analysis and product attractiveness scoring engines.
 - Developed dashboards for retail intelligence and merchandising analytics.
 - Deployed the platform using Docker and cloud deployment platforms such as AWS or Azure.
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3. Architecture Diagram:



4. Modules to be Implemented

1. User Authentication & Role-Based Access

- User registration and login
- JWT authentication
- OAuth2 login
- Role-based access control
- User profile management

Roles

- Store Manager
- Retail Analyst
- Marketing Manager

- Administrator

2. Store & Shelf Management Module

- Store registration
- Shelf mapping
- Product placement management
- Zone management
- Camera assignment

Store Information

- Store ID
- Store Location
- Shelf Layout
- Product Categories
- Camera Locations
- Store Zones

3. Consumer Detection & Tracking Engine

- Shopper detection
- Multi-person tracking
- Path tracking
- Entry/exit monitoring
- Session generation

Detection Features

- Person Detection
- Unique Shopper Tracking
- Path Analysis

- Zone Tracking
- Traffic Monitoring

4. Attention Analysis Engine

- Gaze estimation
- Head pose analysis
- Product attention detection
- Shelf engagement analysis
- Attention duration calculation

Attention Metrics

- Dwell Time
- View Duration
- Shelf Attention Time
- Product Focus Duration
- Repeated Attention Events

5. Product Interaction Analysis Module

- Product pickup detection
- Product return detection
- Shelf interaction monitoring
- Product engagement tracking
- Product comparison analysis

Interaction Events

- Product Viewed
- Product Picked Up
- Product Returned

- Product Purchased
- Product Compared

6. Consumer Behavior Intelligence Engine

- Shopping pattern analysis
- Product preference analysis
- Movement behavior analysis
- Consumer segmentation
- Journey analytics

Consumer Segments

- Explorers
- Quick Buyers
- Comparison Shoppers
- Impulse Buyers
- Brand Loyal Customers

7. Attention Heatmap Generation Engine

- Store heatmap generation
- Shelf heatmap generation
- Product attention heatmaps
- Customer traffic heatmaps
- Engagement hotspot analysis

8. Product Attractiveness Scoring Engine

- Product attractiveness scoring
- Shelf visibility scoring
- Engagement scoring

- Conversion potential scoring
- Marketing effectiveness scoring

Weighted Scoring Model

Product Attractiveness Score =

- Attention Duration (35%)
- Product Interaction Frequency (25%)
- Product Pickup Rate (20%)
- Purchase Conversion Rate (15%)
- Repeat Engagement Rate (5%)

9. Recommendation & Optimization Engine

- Shelf optimization recommendations
- Product placement recommendations
- Promotional placement suggestions
- Consumer engagement recommendations
- Layout improvement suggestions

10. Dashboard & Analytics

Store Manager Dashboard

- Store traffic analytics
- Product engagement insights
- Shelf performance reports
- Conversion metrics

Retail Analyst Dashboard

- Consumer behavior analytics
- Attention heatmaps

- Product attractiveness reports
- Customer journey analytics

Marketing Manager Dashboard

- Campaign effectiveness
- Product visibility analytics
- Promotional performance
- Customer engagement metrics

Admin Dashboard

- User management
- Platform analytics
- Camera management
- System monitoring

11. Notification & Alert System

- Shelf performance alerts
- Product visibility alerts
- Traffic anomaly notifications
- Camera health alerts
- Platform notifications

12. Reports & Export System

- Consumer attention reports
- Product engagement reports
- Shelf performance reports
- Conversion reports
- Marketing effectiveness reports

- PDF export
- Excel export

13. Final Integration, Testing & Deployment

- Frontend and backend integration
 - API validation and testing
 - End-to-end workflow testing
 - Security testing
 - Performance optimization
 - Docker containerization
 - Production deployment
 - Monitoring and logging setup
 - Documentation and user guides
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5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup

Tasks

- Define project objectives and retail intelligence workflows.
- Design system architecture and database schema.
- Create UI wireframes and workflow planning.
- Setup frontend and backend environments.
- Implement authentication and role-based access system.
- Build store and shelf management workflows.
- Integrate retail camera feeds.

Recommended Datasets

Retail Product Checkout Dataset

Purpose:

- Retail object detection
- Product recognition

SKU-110K Dataset

Purpose:

- Shelf product detection
- Retail shelf analytics

COCO Dataset

Purpose:

- Person detection
- Object tracking

Retail Store Traffic Dataset

Purpose:

- Consumer movement analytics
- Store traffic monitoring

Outcomes

- Understand retail attention intelligence workflows.
- Learn system architecture and database design concepts.
- Build frontend and backend project initialization.
- Working authentication and store management system.

Milestone 2: Week 3 & 4 — Consumer Detection & Attention Analysis

Tasks

- Implement shopper tracking engine.
- Build attention analysis workflows.
- Develop gaze estimation models.
- Create dwell time analytics.
- Generate attention reports.

Outcomes

- Consumer tracking engine operational.
- Attention analysis workflows functional.
- Engagement analytics completed.

Milestone 3: Week 5 & 6 — Behavioral Intelligence & Optimization

Tasks

- Implement consumer behavior intelligence engine.
- Build heatmap generation workflows.
- Develop product attractiveness scoring.
- Generate optimization recommendations.
- Create retail intelligence dashboards.

Outcomes

- Consumer behavior engine operational.
- Attention heatmaps functional.
- Optimization recommendation workflows completed.

Milestone 4: Week 7 & 8 — Analytics, Testing & Deployment

Tasks

- Build executive dashboards.

- Add reports and visualization modules.
- Implement testing and validations.
- Deploy platform using Docker and cloud services.
- Prepare final documentation and presentation.

Outcomes

- Fully deployed production-ready platform.
 - Consumer attention intelligence systems operational.
 - Complete end-to-end retail analytics workflow demonstrable.
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6. Evaluation Criteria

Milestone 1 (Week 2)

- Project initialization completed.
- Authentication implemented.
- Store management functional.
- Camera integration operational.

Milestone 2 (Week 4)

- Consumer tracking engine operational.
- Attention analysis workflows functional.
- Gaze estimation implemented.

Milestone 3 (Week 6)

- Behavior intelligence engine operational.
- Heatmap generation functional.
- Recommendation workflows completed.

Milestone 4 (Week 8)

- Fully deployed frontend and backend.
 - Dashboards and reporting systems operational.
 - End-to-end consumer attention workflow demonstrated.
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7. Tools & Tech Stack

Programming Language

Backend

- Python
- FastAPI

Frontend

- JavaScript
- React.js

Database

Primary Database

- PostgreSQL

Secondary Database

- MongoDB

AI & Machine Learning

Computer Vision

- YOLOv8
- OpenCV
- TensorFlow
- PyTorch

- MediaPipe

Behavioral Analytics

- Scikit-learn
- XGBoost
- Pandas
- NumPy

Video Analytics

- OpenCV
- DeepSORT
- ByteTrack
- FFmpeg

Heatmap & Visualization

- Plotly
- Matplotlib
- Seaborn

Streaming & Real-Time Processing

- Kafka
- Redis Streams

Cloud & DevOps

- Docker
- AWS / Azure

Libraries & Frameworks

- FastAPI
- React.js

- Next.js
- Tailwind CSS
- JWT Authentication
- Chart.js

Dev & Deployment Tools

- VS Code
 - Git & GitHub
 - Docker & Docker Compose
 - GitHub Actions
 - Postman
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8. Performance Metrics

Consumer Tracking Metrics

- Person detection accuracy
- Tracking accuracy
- Shopper re-identification accuracy

Attention Analysis Metrics

- Gaze estimation accuracy
- Dwell time accuracy
- Attention mapping precision

Product Engagement Metrics

- Product interaction detection accuracy
- Pickup detection accuracy
- Conversion prediction accuracy

Retail Analytics Metrics

- Shelf optimization effectiveness
- Heatmap accuracy
- Consumer segmentation quality

System Performance Metrics

- Video processing latency
 - API response time
 - Dashboard loading speed
 - Concurrent camera handling capacity
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9. Example Quantitative Goals

Consumer Attention Analysis

Accurately identify which products and shelves receive the highest customer attention.

Product Engagement Intelligence

Measure customer interactions and product attractiveness effectively.

Retail Optimization

Improve product placement and merchandising decisions through AI-driven insights.

Consumer Behavior Analytics

Understand shopping journeys and engagement patterns across retail environments.

Platform Performance

Support real-time retail video analytics and consumer intelligence workflows while maintaining stable performance and responsiveness.